

Measures of Variations (Dispersions)

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Empirical Rules

Empirical rule

The empirical rule states that for a *normal distribution* (or a *bell-shaped distribution*), nearly all of the data will fall within *three standard deviations* of the mean. The empirical rule can be broken down into three parts:

Empirical rule

- Approximately 68% of data falls within the *first* standard deviation from the mean.
- Approximately 95% fall within *two* standard deviations.
- Approximately 99.7% fall within *three* standard deviations.

The rule is also called the *68-95-99.7 Rule* or the *Three Sigma Rule*.

Empirical rule

When applying the Empirical Rule to a data set the following conditions are true:

Approximately **68%** of the data falls within **one** standard deviation of the mean (or between the mean – 1 times the standard deviation, and the mean + 1 times the standard deviation). In mathematical notation, this is represented as: **$\mu \pm 1\sigma$**

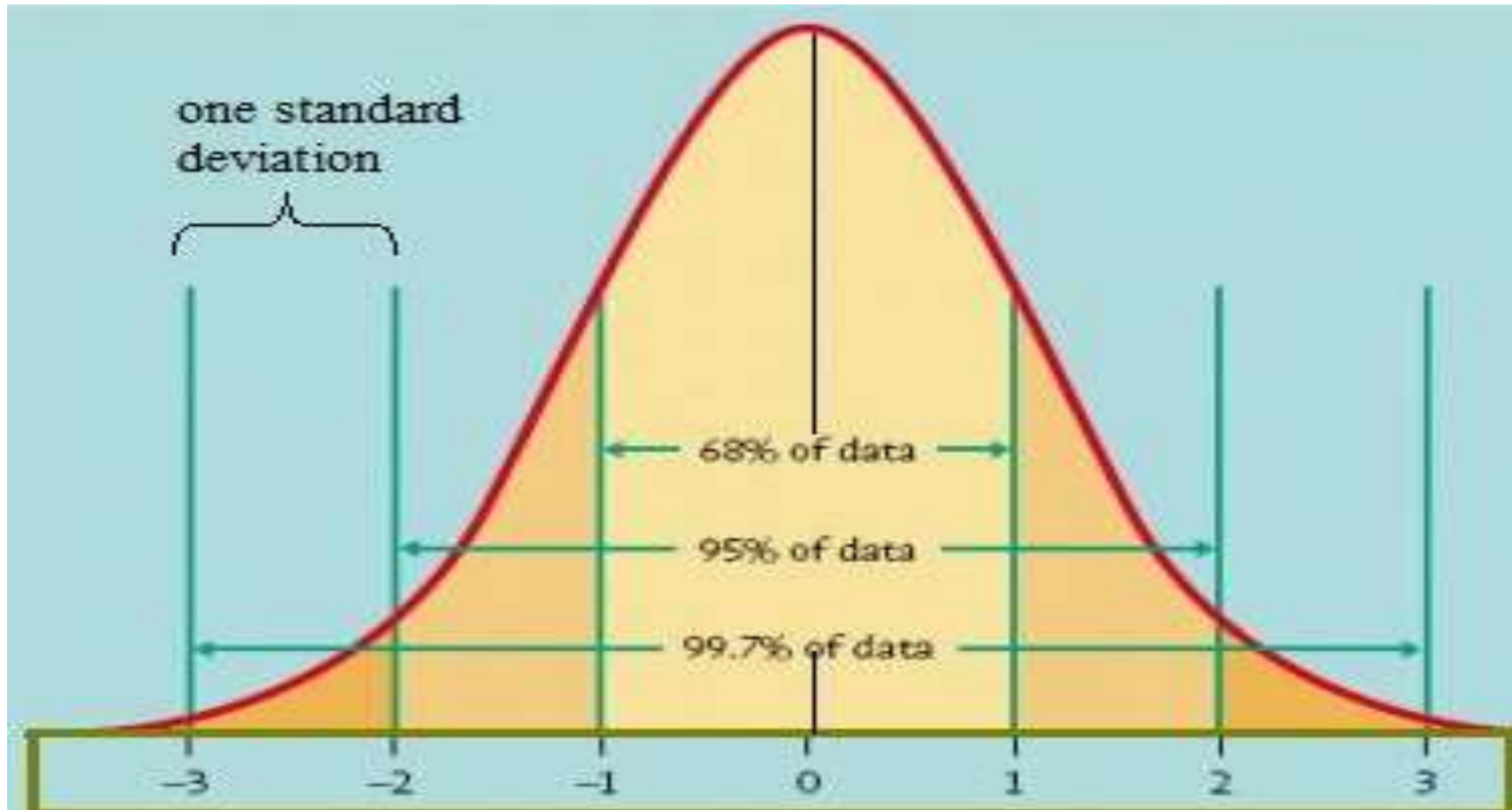
Empirical rule

- Approximately **95%** of the data falls within **two** standard deviations of the mean (or between the mean – 2 times the standard deviation, and the mean + 2 times the standard deviation). The mathematical notation for this is: **$\mu \pm 2\sigma$**

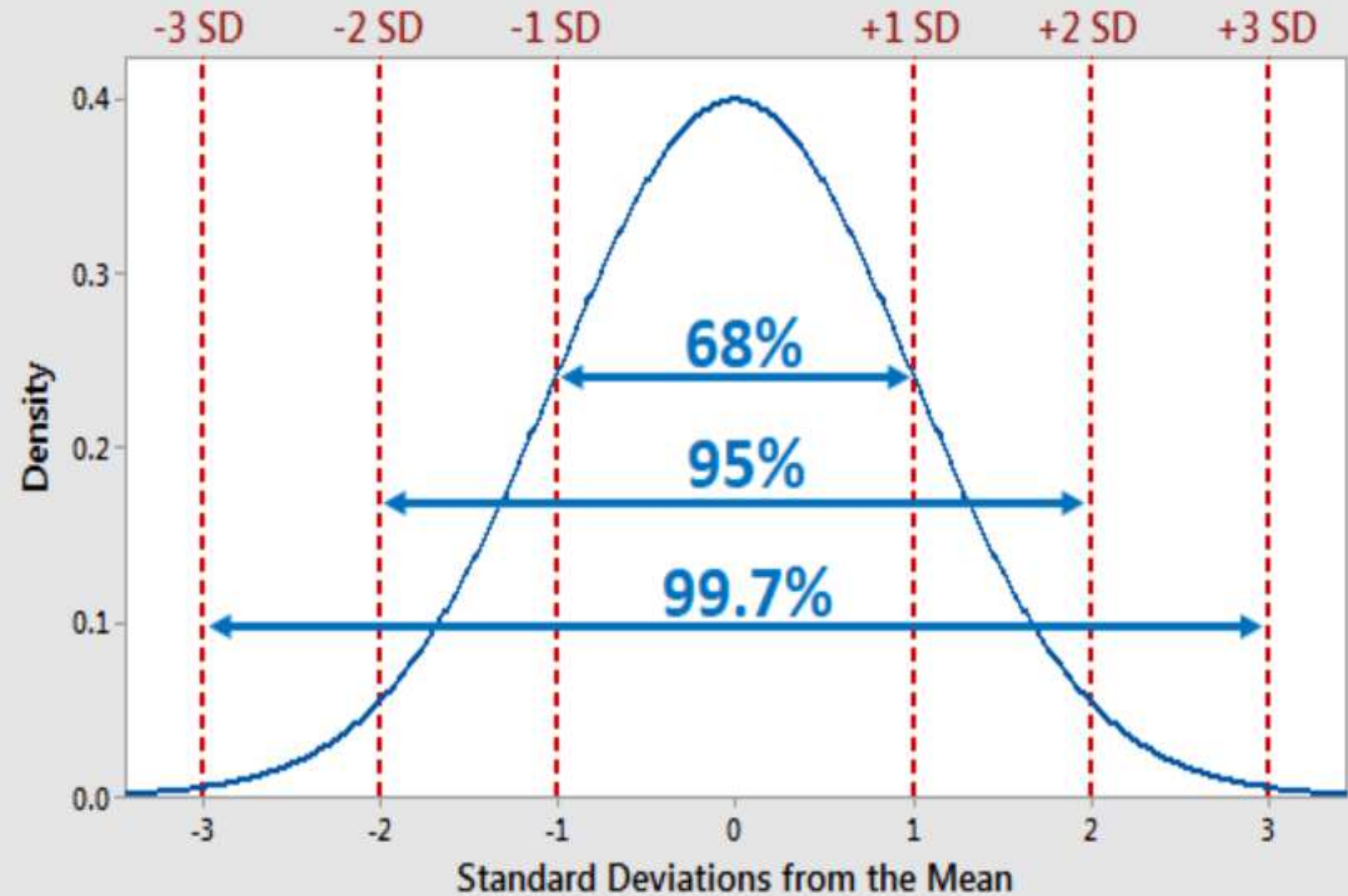
Empirical rule

- Approximately **99.7%** of the data falls within **three** standard deviations of the mean (or between the mean – 3 times the standard deviation and the mean + 3 times the standard deviation). The following notation is used to represent this fact: **$\mu \pm 3\sigma$**

Empirical rule



Chebyshev's Theorem



Empirical rule

What is the Empirical Rule?

- The empirical rule, also referred to as the three-sigma rule or **68-95-99.7 rule**, is a statistical rule which states that for ***a normal distribution***, almost all data falls within three standard deviations (denoted by σ) of the mean (denoted by μ).

Empirical rule

- Broken down, the empirical rule shows that 68% falls within the first standard deviation ($\mu \pm \sigma$), 95% within the first two standard deviations ($\mu \pm 2\sigma$), and 99.7% within the first three standard deviations ($\mu \pm 3\sigma$).

Empirical rule

Example10:- Assume that the distribution an animal in the zoo lives is bell shaped with average of 10 years and standard deviation of 1.4 years. The zookeeper attempts to figure out the probability of an animal living for more than 7.2 years. This distribution looks as follows:

Empirical rule

- One standard deviation ($\mu \pm \sigma$): 8.6 to 11.4 years
- Two standard deviations ($\mu \pm 2\sigma$): 7.2 to 12.8 years
- Three standard deviations ($\mu \pm 3\sigma$): 5.8 to 14.2 years

The empirical rule states that approximately 95% of the distribution lies within two standard deviations.

Empirical rule

Thus, 5% lies outside of two standard deviations; half above 12.8 years and half below 7.2 years. Thus, the probability of living for more than 7.2 years is:

$$95\% + (5\% / 2) = 97.5\%$$

Empirical rule

Example10:- Let's assume a population of animals in a zoo is known to be normally distributed. Each animal lives 13.1 years on average (mean), and the standard deviation of the lifespan is 1.5 years.

Empirical rule

If someone wants to know the probability that an animal will live longer than 14.6 years, they could use the empirical rule. Knowing the distribution's mean is 13.1 years old, the following age ranges occur for each standard deviation:

Empirical rule

- One standard deviation ($\mu \pm \sigma$): $(13.1 - 1.5)$ to $(13.1 + 1.5)$, or 11.6 to 14.6
- Two standard deviations ($\mu \pm 2\sigma$): $13.1 - (2 \times 1.5)$ to $13.1 + (2 \times 1.5)$, or 10.1 to 16.1
- Three standard deviations ($\mu \pm 3\sigma$): $13.1 - (3 \times 1.5)$ to $13.1 + (3 \times 1.5)$, or, 8.6 to 17.6

Empirical rule

The person solving this problem needs to calculate the total probability of the animal living 14.6 years or longer. The empirical rule shows that 68% of the distribution lies within one standard deviation, in this case, from 11.6 to 14.6 years.

Empirical rule

Thus, the remaining 32% of the distribution lies outside this range. One half lies above 14.6 and the other below 11.6. So, the probability of the animal living for more than 14.6 is 16% (calculated as 32% divided by two).

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The Standard Score (The Z - Score)

The Standard Score (Z-Score)

Definition 5:- The variable

$$Z_p = \frac{x - \mu}{\sigma} \quad (\text{ or } Z_s = \frac{x - \bar{x}}{s}) \text{ which measures}$$

the deviation from the mean in units of standard deviation is called **a standard score (or a Z-score).**

The Standard Score (Z-Score)

The purpose of *Z-score* or *standard score* is to identify and describe the *exact location of every score* in a distribution.

The Altman Z-Score Formula

- The Altman Z-score is the output of a **credit-strength test** that helps gauge the likelihood of **bankruptcy** for a publicly traded manufacturing company. The Z-score is based on five key financial ratios that can be found and calculated from a company's annual report. The calculation used to determine the Altman Z-score is as follows:

The Altman Z-Score Formula

The original Z-score formula was as follows:

$$\zeta = 1.2A + 1.4B + 3.3C + 0.6D + 1.0E$$

where: **Zeta(ζ)** =The Altman Z-score

A=Working capital/total assets

B=Retained earnings/total assets

C=Earnings before interest and taxes (EBIT) /total assets

D=Market value of equity/book value of total liabilities

E=Sales/total assets

The Altman Z-Score Formula

Edward Altman, a professor at New York University, developed and introduced the Z-score formula in the late 1960s as a solution to the time-consuming and somewhat confusing process investors had to undergo *to determine how close to bankruptcy a company was.*

The Altman Z-Score Formula

In addition to this, Z-scores also make it possible for analysts to adapt scores from various data sets to make scores that can be compared to one another accurately. *Usability testing* is one example of a real-life application of Z-scores.

The Altman Z-Score Formula

Why It's Useful in Usability Testing:

- **Compare Across Users or Tasks**

Helps you see if a user performed **better or worse than average**.

- **Normalize Different Metrics**

Makes it easier to compare **task completion times**, **SUS scores**, or **error rates** on the same scale.

The Altman Z-Score Formula

- **Identify Outliers**

A high or low z-score shows performance that is significantly **different from the norm**, indicating potential usability issues or standout success.

The Altman Z-Score Formula

Example:

Suppose the average task completion time is **60 seconds**, with a standard deviation of **10 seconds**. A user who finishes in **40 seconds**:

$$Z = \frac{40 - 60}{10} = -2$$

This means the user was **2 standard deviations faster** than average — a strong indicator of good usability

The Standard Score (Z-Score)

- A positive ***z-score*** indicates the raw score is higher than the mean average. For example, if a ***z-score*** is equal to +1, it is 1 standard deviation **above** the mean. A negative ***z-score*** reveals the raw score is below the mean average. For example, if a ***z-score*** is equal to -2 , it is 2 standard deviations **below** the mean.

The Standard Score (Z-Score)

- Simply put, a **z-score** (also called a **standard score**) gives you an idea of how far from the mean a data point is. But more technically it's a measure of how many standard deviations below or above the population (sample) mean a raw score is.

The Standard Score (Z-Score)

Example1:- Scores on a history test have average of 80 with standard deviation of 6. What is the z-score for a student who earned a 75 on the test?

Solution:- Given $\bar{x} = 80$, $s = 6$, $x = 75$ and required Z.

We know that $Z_p = \frac{x - \mu}{\sigma}$ (or $Z_s = \frac{x - \bar{x}}{s}$)

$$\text{Thus } Z = \frac{75 - 80}{6} = \frac{-5}{6} = -0.833$$

The Standard Score (Z-Score)

Example2:- The weight of chocolate bars from a particular chocolate factory has a mean of 8 ounces with standard deviation of 0.1 ounce. What is the z-score corresponding to a weight of 8.17 ounces?

Solution:- Given $\bar{x} = 8$, $\sigma = 0.1$, $x = 8.17$ and required Z.

We know that $Z_p = \frac{x - \mu}{\sigma}$ (or $Z_s = \frac{x - \bar{x}}{s}$)

$$\text{Thus } Z = \frac{8.17 - 8}{0.1} = \frac{0.17}{0.1} = 1.7$$

The Standard Score (Z-Score)

Example3:- The temperature is recorded at 60 airports in a region. The average temperature is 67 degrees Fahrenheit with standard deviation of 5 degrees. What is the z-score for a temperature of 68 degrees?

Example4:- A group of friends compares what they received while trick or treating. They find that the average number of pieces of candy received is 43, with standard deviation of 2. What is the z-score corresponding to 20 pieces of candy?

The Standard Score (Z-Score)

Example5:- A particular leg bone for dinosaur fossils has a mean length of 5 feet with standard deviation of 3 inches. What is the z-score that corresponds to a length of 62 inches?

Example6:- Books in the library are found to have average length of 350 pages with standard deviation of 100 pages. What is the z-score corresponding to a book of length 80 pages?

The Standard Score (Z-Score)

Z-scores are often used in academic settings to analyze how well a student's score compares to the mean score on a given exam.

Example 7:- Suppose the scores on a certain college entrance exam are roughly normally distributed with a mean of 82 and a standard deviation of 5. If a certain student received a 90 on the exam, find the standard score and interpret your result.

The Standard Score (Z-Score)

Z-scores are often used in a medical setting to analyze how a certain newborn's weight compares to the mean weight of all babies.

Example 8:- It's well-documented that the weights of newborns are normally distributed with a mean of about 7.5 pounds and a standard deviation of 0.5 pounds. If a certain newborn weighs 7.7 pounds, calculate the standard score and interpret your result.

The Standard Score (Z-Score)

Z-scores are often used in a biology to assess how the height of a certain animal compares to the mean population height of that particular animal.

Example 9:- Suppose the heights of a certain species of giraffe is normally distributed with a mean of 16 feet and a standard deviation of 2 feet. If a certain giraffe from this species is 15 feet tall, calculate the z-score and interpret your result.

The Standard Score (Z-Score)

Z-scores are often used in medical settings to assess how an individual's blood pressure compares to the mean population blood pressure.

Example 10:- The distribution of diastolic blood pressure for men is normally distributed with a mean of about 80 and a standard deviation of 20. If a certain man has a diastolic blood pressure of 100, calculate the z-score and interpret your result.

The Standard Score (Z-Score)

Blood pressure is measured using two numbers:
The first number, called **systolic** blood pressure, measures the pressure in your arteries when your heart beats. The second number, called **diastolic** blood pressure, measures the pressure in your arteries when your heart rests between beats.

The Standard Score (Z-Score)

Z-scores allow you to take data points drawn from populations with different means and standard deviations and place them on a common scale. This standard scale lets you compare observations for different types of variables that would otherwise be difficult.

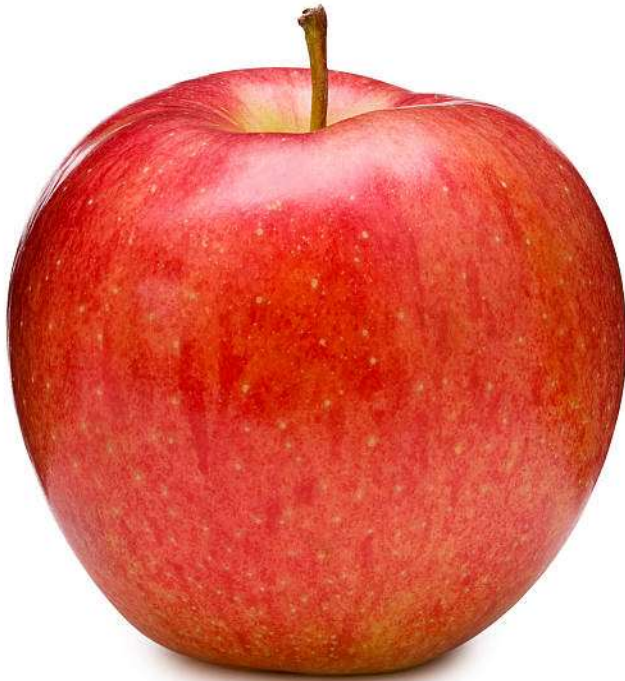
The Standard Score (Z-Score)

That's why z-scores are also known as standard scores, and the process of transforming raw data to z-scores is called standardization. It lets you compare data points across variables that have different distributions.

Imagine we literally need to compare apples to oranges. Specifically, we'll compare their weights.

The Standard Score (Z-Score)

We have a 110-gram apple and a 100-gram orange.



The Standard Score (Z-Score)

By comparing the raw values, it's easy to see the apple weighs slightly more than the orange. However, let's compare their z-scores. To do this, we need to know the means and standard deviations for the populations of apples and oranges.

The Standard Score (Z-Score)

Assume that apples and oranges follow a normal distribution with the following properties:

	Apple	Orange
Average weights (in grams)	100	140
Standard deviation	15	25

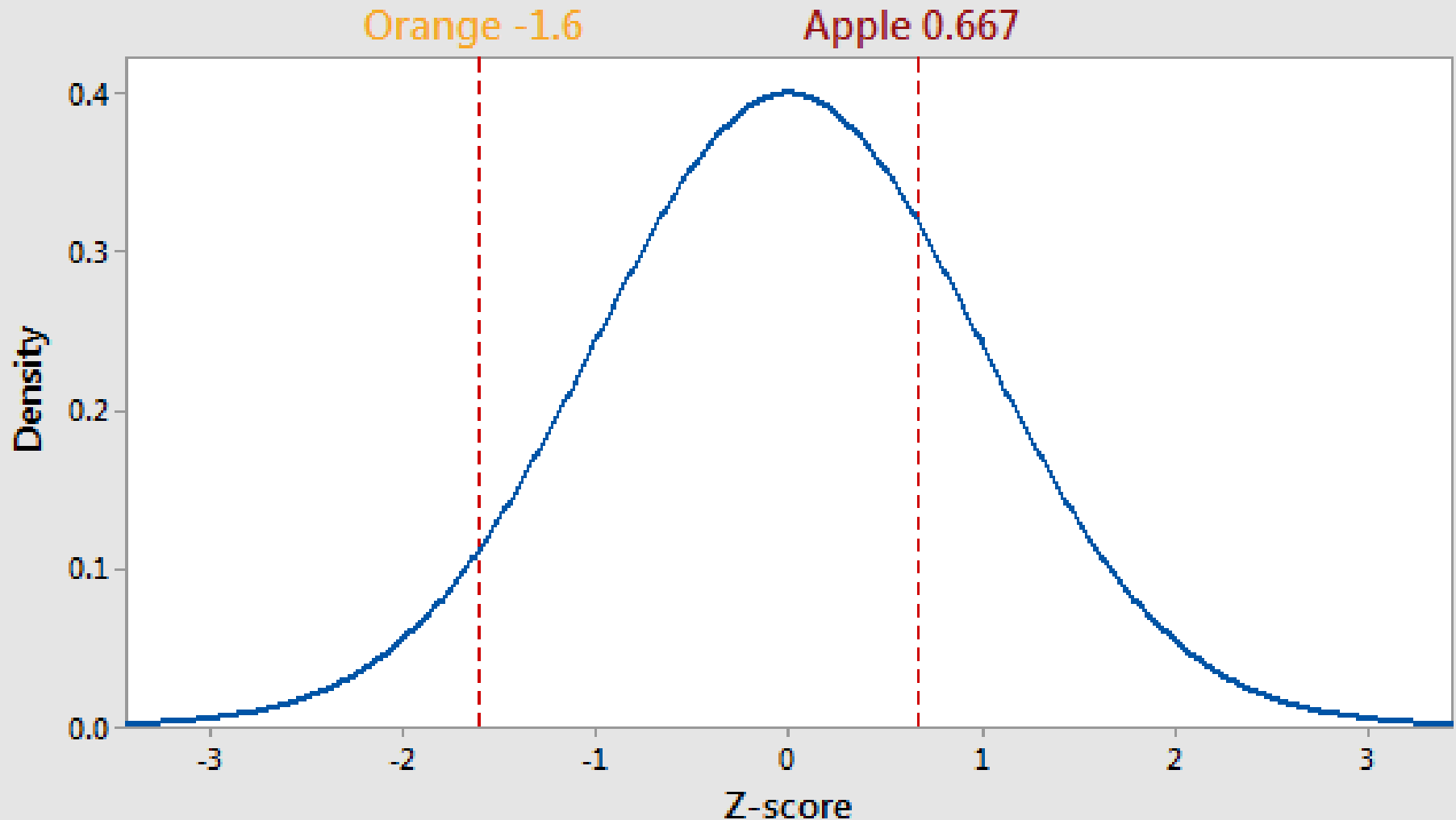
The Standard Score (Z-Score)

The apple's positive z-score (0.667) signifies that it is heavier than the average apple. It's not an extreme value, but it is above the mean. Conversely, the orange has a markedly negative Z-score (-1.6). It's well below the mean weight for oranges. Let's see these standard scores in the standard normal distribution in the next slide:

The Standard Score (Z-Score)

Comparing Apples to Oranges Using Z-scores

Normal, Mean=0, StDev=1



The Standard Score (Z-Score)

Our apple is a bit heavier than average, while the orange is small! Using z-scores, we learned where each fruit falls within its distribution and how they compare.

The Standard Score (Z-Score)

Example 11:- Suppose Teka received a score $x = 60$ on Operating System course and a score of $x = 56$ on Networking course exams. For which course should Teka expect the better grade?

The Standard Score (Z-Score)

The information given was not enough.

Example 12:- Suppose Teka received a score $x = 60$ on final exam in Operating System course for which the average mark was 55 and the standard deviation was 8. On the final exam in Networking course in which the average mark was 51 and the standard deviation was 10 he received a score of $x = 56$. For which course was Teka's relative standing higher?

The Standard Score (Z-Score)

Example 13:- The state college requires applicants to submit test scores from either the College Placement Exam (CPE) or the College Board Test (CBT). Scores on the CPE have a mean of $\mu = 200$ with $\sigma = 50$ and scores on the CBT averages $\mu = 500$ with $\sigma = 100$. Sadat's application includes a CPE score of $x = 235$ and Dagmawit's application reports a CBT score of $x = 540$. Based on these scores, which student is more likely to be admitted?

The Standard Score (Z-Score)

Example 14:- Heights for traditional college-age students in the US have respective means and standard deviations of approximately 70 inches and 3 inches for males and 165.1 cm and 6.35 cm for females. If a male college student were 68 inches tall and a female college student was 160 cm tall, who is relatively shorter in their respected gender groups?

Any Questions



Any Questions

End of Chapter 5

